

ECON 101: Introduction to Economics

Lecture 2: Supply and Demand

1 Motivation

Last lecture we established that economics is about allocating scarce resources. *How* does a decentralised economy coordinate those decisions? The answer, in most markets, is the **price system**: prices rise and fall until the quantity sellers want to sell equals the quantity buyers want to buy. Supply and demand is the workhorse model for understanding that process.

2 Demand

Definition 2.1. The **quantity demanded** of a good is the amount buyers are willing and able to purchase at a given price, holding all other factors constant (*ceteris paribus*).

Definition 2.2. The **Law of Demand**: as the price of a good rises, the quantity demanded falls; as the price falls, the quantity demanded rises. All else equal, $P \uparrow \Rightarrow Q^d \downarrow$.

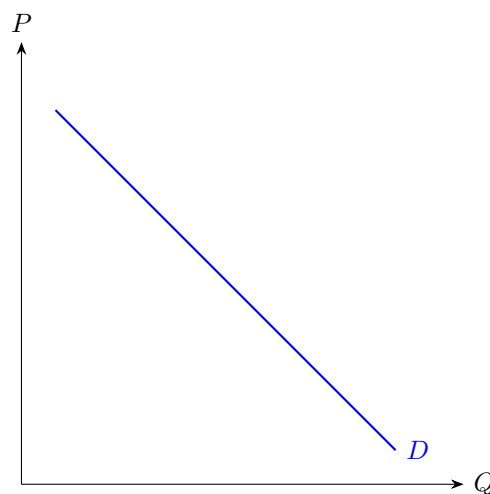
Two complementary reasons underlie the law of demand:

- **Substitution effect**: a higher price makes other goods relatively cheaper, so buyers substitute away.
- **Income effect**: a higher price erodes real purchasing power, reducing consumption of normal goods.

2.1 The Demand Curve

A **demand curve** plots the quantity demanded at each price. It slopes downward by the law of demand. A linear example:

$$Q^d = a - bP, \quad a, b > 0.$$



2.2 Shifters of Demand

A *change in price* causes a **movement along** the demand curve. A *change in any other determinant* causes a **shift** of the entire curve. The main shifters are:

- (1) **Income.** For a *normal good*, higher income $\Rightarrow$ demand shifts right. For an *inferior good*, it shifts left.
- (2) **Prices of related goods.**
 - *Substitutes* (e.g. coffee and tea): $P_{\text{coffee}} \uparrow \Rightarrow D_{\text{tea}}$ shifts right.
 - *Complements* (e.g. coffee and cream): $P_{\text{coffee}} \uparrow \Rightarrow D_{\text{cream}}$ shifts left.
- (3) **Tastes and preferences.** Advertising, trends, new information.
- (4) **Expectations.** If buyers expect a future price rise, current demand increases.
- (5) **Number of buyers.** More buyers $\Rightarrow$ market demand shifts right.

Example 2.3. A health report links red meat to cardiovascular disease. Tastes shift: demand for beef falls (curve shifts left) and demand for chicken rises (curve shifts right), even though neither price changed.

3 Supply

Definition 3.1. The **quantity supplied** is the amount sellers are willing and able to offer for sale at a given price, *ceteris paribus*.

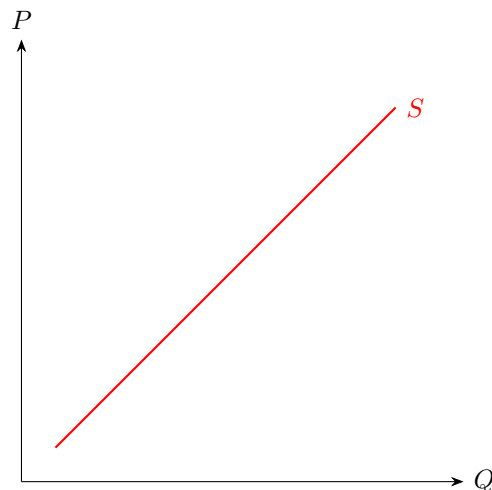
Definition 3.2. The **Law of Supply**: as price rises, quantity supplied rises; as price falls, quantity supplied falls. $P \uparrow \Rightarrow Q^s \uparrow$.

The intuition: a higher price raises the profitability of production, inducing existing firms to expand output and attracting new entrants.

3.1 The Supply Curve

A **supply curve** slopes upward. A linear example:

$$Q^s = -c + dP, \quad c, d > 0, \quad P \geq c/d.$$



3.2 Shifters of Supply

- (1) **Input prices.** Cheaper inputs $\Rightarrow$ supply shifts right.
- (2) **Technology.** Improved technology lowers production costs and shifts supply right.
- (3) **Expectations.** Sellers who expect higher future prices may withhold supply today (shift left now).
- (4) **Number of sellers.** More producers $\Rightarrow$ market supply shifts right.
- (5) **Government policies.** Taxes raise costs (shift left); subsidies lower costs (shift right).

4 Market Equilibrium

Definition 4.1. A market is in **equilibrium** when quantity demanded equals quantity supplied:

$$Q^d(P^*) = Q^s(P^*).$$

The price P^* is the **equilibrium price** (market-clearing price) and $Q^* \equiv Q^d(P^*)$ is the **equilibrium quantity**.

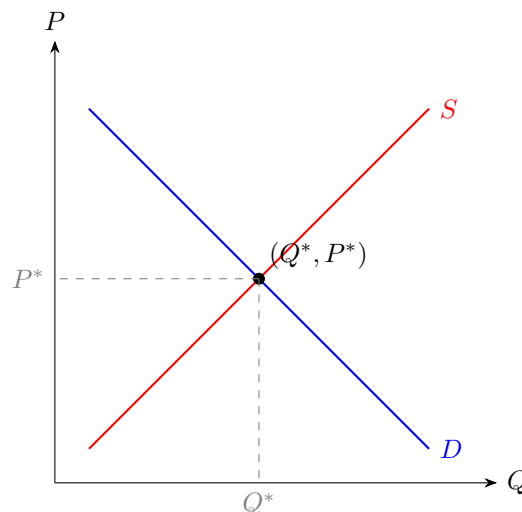
4.1 Solving for Equilibrium

Given

$$Q^d = a - bP \quad \text{and} \quad Q^s = -c + dP,$$

set $Q^d = Q^s$:

$$a - bP^* = -c + dP^* \implies P^* = \frac{a + c}{b + d}, \quad Q^* = a - b \cdot \frac{a + c}{b + d} = \frac{ad - bc}{b + d}.$$



4.2 Surplus and Shortage

- **Surplus** ($P > P^*$): $Q^s > Q^d$. Unsold inventory accumulates; sellers cut prices, restoring equilibrium.
- **Shortage** ($P < P^*$): $Q^d > Q^s$. Buyers bid prices up, restoring equilibrium.

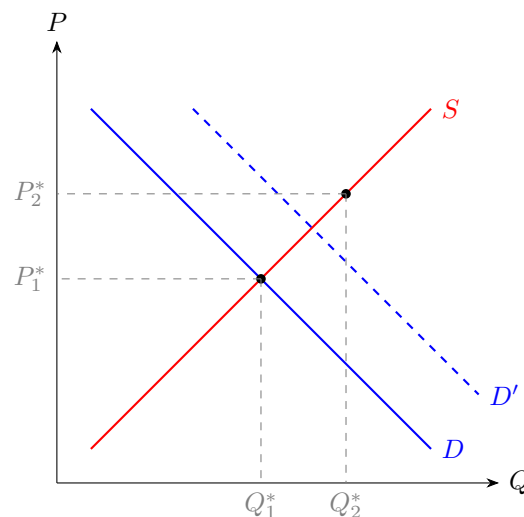
The market mechanism self-corrects: deviations from P^* generate pressures that push prices back toward equilibrium.

5 Comparative Statics

Comparative statics asks: *if one determinant changes, how do P^* and Q^* change?* The procedure is:

- (1) Determine whether the event shifts D , S , or both, and in which direction.
- (2) Draw the new curve.
- (3) Read off the change in equilibrium.

Example 5.1 (Demand increase). A rise in consumer income (normal good) shifts D rightward to D' , raising both P^* and Q^* .



Example 5.2 (Supply decrease). An increase in input costs shifts S leftward to S' , raising P^* but lowering Q^* .

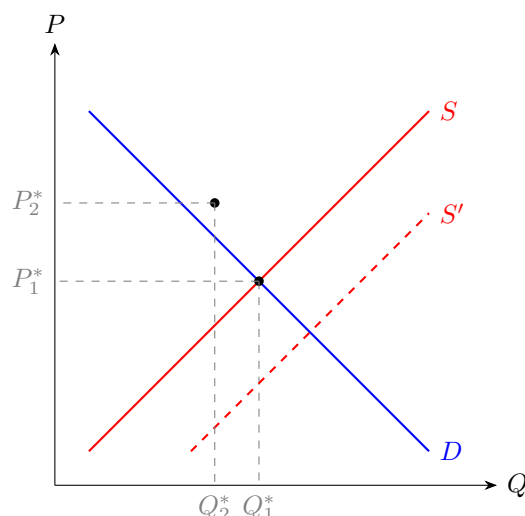


Table 1 summarises all four single-shift cases.

Table 1: Comparative statics: direction of change in equilibrium

Event	ΔP^*	ΔQ^*
Demand increases (D shifts right)	+	+
Demand decreases (D shifts left)	−	−
Supply increases (S shifts right)	−	+
Supply decreases (S shifts left)	+	−

Remark. When *both* curves shift simultaneously, the direction of one of the two equilibrium variables becomes ambiguous without knowing the magnitudes of the shifts.

6 Applications: Price Controls

Governments sometimes prevent prices from reaching P^* .

Definition 6.1. A **price ceiling** is a legal maximum price. It is binding only when set below P^* , creating a **shortage**: $Q^d > Q^s$.

A **price floor** is a legal minimum price. It is binding only when set above P^* , creating a **surplus**: $Q^s > Q^d$.

Example 6.2 (Rent control). A city caps apartment rent below the market-clearing level (a binding price ceiling). Quantity demanded of apartments exceeds quantity supplied, producing a housing shortage. Non-price rationing mechanisms (waiting lists, connections) emerge.

Example 6.3 (Minimum wage). A binding minimum wage sets the wage floor above the market wage. Quantity of labour supplied exceeds quantity demanded, producing unemployment.

7 Summary

- The **law of demand**: $P \uparrow \Rightarrow Q^d \downarrow$ (downward-sloping demand curve).

- The **law of supply**: $P \uparrow \Rightarrow Q^s \uparrow$ (upward-sloping supply curve).
- **Equilibrium** ($Q^d = Q^s$) determines P^* and Q^* . Surpluses and shortages push prices back to P^* .
- Shifts in D or S change equilibrium; use comparative statics to determine the direction.
- **Price controls** prevent market clearing, generating persistent shortages (ceilings) or surpluses (floors).